EXPERIMENT – 02

AIM:

To write JavaScript code to validate user input in a registration form. The validation will ensure that the data entered by the user meets specific criteria for each field. This will help in maintaining data integrity and prevent invalid submissions.

ALGORITHM:

1. Input:

The user fills in the following fields: First Name, Password, E-mail, Mobile Number, Last Name, and Address.

2. Validation Steps:

1. Validate First Name:

o Check if the first name contains only alphabets.

o Ensure the length of the First Name is at least 6 characters.

o If any of these conditions fail, display an error message indicating the required input.

2. Validate Password:

o Check if the password is at least 6 characters long.

o If it is less than 6 characters, display an error message.

3. Validate E-mail:

o Use a regular expression to check if the email follows the standard pattern: name@domain.com.

o Ensure that the email contains no invalid characters and matches the required pattern.

o If it doesn't match, display an error message indicating an invalid email format.

4. Validate Mobile Number:

o Check if the mobile number contains exactly 10 digits.

o Ensure there are no alphabets or special characters.

o If the number is invalid, display an error message indicating the correct format.

5. Validate Last Name:

o Ensure the Last Name field is not empty.

o If it is empty, display an error message.

6. Validate Address:

o Ensure the Address field is not empty.

o If it is empty, display an error message.

PROGRAM:

HTML File:

```
<!DOCTYPE html>

<html>

<head>

<title>Hostel Registration</title>

<style>

body {

  margin: 0;

  height: 100vh;

  background-color: #1a1a2e;

  font-family: Arial, Helvetica, sans-serif;

  display: flex;

  justify-content: center;

  align-items: center;

}


.form-container {

  background: rgba(95, 91, 91, 0.95);

  padding: 30px 40px;

  border-radius: 12px;

  box-shadow: 0 8px 20px rgba(0,0,0,0.4);

  width: 380px;

}
```

```
h2 {
  text-align: center;
  margin-bottom: 5px;
  color: #fff;
}

.subtitle {
  text-align: center;
  color: #ccc;
  font-size: 13px;
  margin-bottom: 20px;
}

table {
  width: 100%;
}

td {
  padding: 8px 0;
  color: #ddd;
  font-size: 14px;
}

input, textarea, select {
  width: 100%;
  padding: 8px;
```

```
  border-radius: 6px;
  border: 1px solid #ccc;
  font-size: 14px;
  box-sizing: border-box;
}

textarea {
  resize: none;
}

button {
  width: 100%;
  padding: 10px;
  margin-top: 10px;
  background-color: #4CAF50;
  color: white;
  border: none;
  border-radius: 6px;
  font-size: 16px;
  cursor: pointer;
}

button:hover {
  background-color: #45a049;
}
</style>
</head>
```

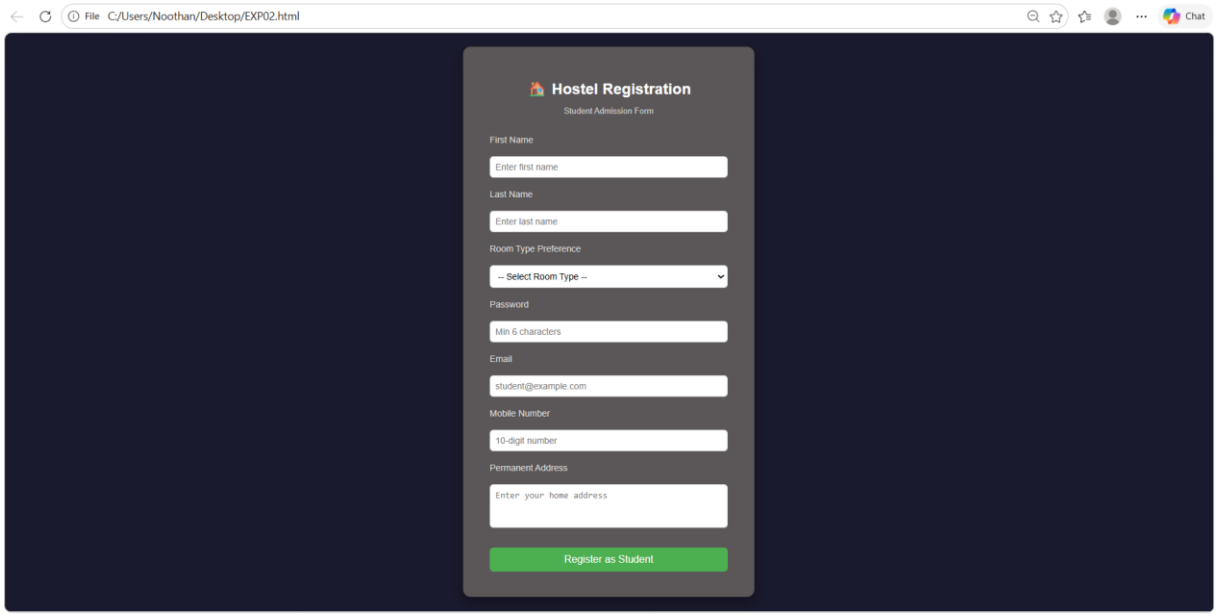
```
<body>
<div class="form-container">
  <h2>🏠 Hostel Registration</h2>
  <p class="subtitle">Student Admission Form</p>

  <form onsubmit="return validateForm()" action="#">
    <table>
      <tr><td>First Name</td></tr>
      <tr><td><input type="text" id="fname" placeholder="Enter first name"></td></tr>

      <tr><td>Last Name</td></tr>
      <tr><td><input type="text" id="lname" placeholder="Enter last name"></td></tr>

      <tr><td>Room Type Preference</td></tr>
      <tr>
        <td>
          <select id="roomtype">
            <option value="">-- Select Room Type --</option>
            <option value="single">Single Occupancy</option>
            <option value="double">Double Occupancy</option>
            <option value="triple">Triple Occupancy</option>
          </select>
        </td>
      </tr>

      <tr><td>Password</td></tr>
```

```
    <tr><td><input type="password" id="password" placeholder="Min 6 characters"></td></tr>


    <tr><td>Email</td></tr>

    <tr><td><input type="text" id="email" placeholder="student@example.com"></td></tr>


    <tr><td>Mobile Number</td></tr>

    <tr><td><input type="text" id="mobile" placeholder="10-digit number"></td></tr>


    <tr><td>Permanent Address</td></tr>

    <tr><td><textarea id="address" rows="3" placeholder="Enter your home
address"></textarea></td></tr>


    <tr><td><button type="submit">Register as Student</button></td></tr>

   </table>

  </form>

</div>


<script>

function validateForm() {

  var fname    = document.getElementById("fname").value.trim();

  var lname    = document.getElementById("lname").value.trim();

  var roomtype = document.getElementById("roomtype").value;

  var password = document.getElementById("password").value;

  var email    = document.getElementById("email").value.trim();

  var mobile   = document.getElementById("mobile").value.trim();

  var address  = document.getElementById("address").value.trim();


  var namePattern = /^[A-Za-z]{3,}$/;
```

```
if (!namePattern.test(fname)) {
  alert("First Name must contain only alphabets and minimum 3 characters");
  return false;
}

if (lname === "") {
  alert("Last Name should not be empty");
  return false;
}

if (roomtype === "") {
  alert("Please select a Room Type preference");
  return false;
}

if (password.length < 6) {
  alert("Password must be at least 6 characters long");
  return false;
}

var emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
if (!emailPattern.test(email)) {
  alert("Please enter a valid Email ID");
  return false;
}

var mobilePattern = /^[0-9]{10}$/;
```

```
  if (!mobilePattern.test(mobile)) {
    alert("Mobile Number must contain exactly 10 digits");
    return false;
  }

  if (address === "") {
    alert("Permanent Address should not be empty");
    return false;
  }

  alert("Student Registration Successful! Welcome to the Hostel.");
  return true;
}
</script>
</body>
</html>
```

OUTPUT:



Result: HTML page with Form is created and is validated using Javascript.